Chemical Diffusion Training App

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Context

Microenvironment

- Domain Size
- Voxel Size
- Substrates
- Boundary Conditions

Cells

- Custom Data
- Cell Functions
- Phenotype
- Cell State
- Basic Agent Variables(inherited)
- Cell Definition

Global Variables

- User Parameters
- PhysiCell Constants
- List of cells
- SVG options
- MultiCellDS options
- Default Cell Definition
- Default microenvironment

Background

- Cells have different chemical molecules and substrates in their environment. These environmental factors impact how cells and tissues grow.
- PhysiCell includes certain parameters to implement the cells' environment, or the **microenvironment**.
- Microevironment is the stage for cells to act and interact.

Mathematics: Variables and Definitions

- **Domain Size (in microns)**
  - ✦the dimensions for the area in which the simulation takes place
- **Voxel Size**
  - ✦Domain is divided into a mesh to carry out diffusion
  - ✦Voxel size influences how quickly chemicals can diffuse across the domain.
  - ✦If a chemical substrate exists in a voxel, it occupies the entire area of the voxel.
    - ♦ larger and fewer voxels: decrease cost in computation time but more difficult to detect patterns
    - ♦ smaller voxels: diffusion patterns are more detailed due to fine grained grid but computational cost will increase
- **Diffusion Rate:** Each substrate has its own diffusion rate
- **Decay Rate:** Each substrate has its own decay rate

Mathematics: Variables and Definitions(cont.)

- **Boundary Conditions**
  - ✦Behavior of chemical substances at the edges of domain
    - ♦ **Neumann Boundary Conditions**
      - » Chemicals substances bounce off the edge of domain
    - ♦ **Dirichlet Boundary Conditions**
      - » Domain boundaries can act as source or sink
      - » It is possible to make some voxels to act as Dirichlet nodes

Mathematics: Models

Movement of chemical substances in microenvironment is controlled by following equation

- $\boldsymbol{\rho}$ is vector of substrates
- $\boldsymbol{\rho}^*$ is the vector of saturation densities
- **D** is vector of diffusion rates
- $\boldsymbol{\lambda}$ is vector of decay rates
- **S** is bulk supply rate
- **U** is the uptake function

$$\frac{\partial \rho}{\partial t} = \overbrace{\mathbf{D}\nabla^2\boldsymbol{\rho}}^{\text{diffusion}} - \overbrace{\boldsymbol{\lambda}\boldsymbol{\rho}}^{\text{decay}} + \overbrace{\mathbf{S}(\boldsymbol{\rho}^* - \boldsymbol{\rho})}^{\text{bulk source}} - \overbrace{\mathbf{U}\boldsymbol{\rho}}^{\text{bulk uptake}}$$

$$+ \overbrace{\sum_{\text{cells } k} \delta(\mathbf{x} - \mathbf{x}_k) W_k \left[\mathbf{S}_k(\boldsymbol{\rho}_k^* - \boldsymbol{\rho}) - \mathbf{U}_k\boldsymbol{\rho}\right]}^{\text{sources and uptake by cells}} \quad \text{in } \Omega$$

Demonstration

- In this app, two substrates are simulated in a microenvironment :
  - ✦Oxygen and Glucose.

- There is an option for you to make one voxel in the domain as a Dirichlet node and set the concentration of substrates there.

Class structure: Data

- Microenvironment can be controlled mainly through XML configuration file (recommended)
- But you can also edit it through *setup_microenvironment* function in you custom.cpp
- Mostly *setup_microenvironment* function is used to handle some extra features that are not readily available in xml

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